Improving Mathematical Reasoning in LLMs via Formal Specification Injection

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Can Machines Think?

Alan Turing posed this question in 1950 *"Computing Machinery and Intelligence"*.

- A human interrogator communicates with an agent through text alone and must determine whether it is a human or a machine.
- This mirrors how modern LLMs are deployed: users interact entirely through a text interface.
- Mathematical problem solving provides a natural test: it requires not only language comprehension but **structured multi-step reasoning**.

Human

Benny saw a **10**-ft shark with **2 6**-inch remoras. What % of the shark's length is the combined remora length?

Large Language Model

Let's think step by step.
Combined remora length = 2 × 6 = 12 inches.
Shark length = 10 × 12 = 120 inches.
12 / 120 × 100 = **10%**. The final answer is **10**.

- But what if the **numbers** and **names** are changed? Can the machine still answer correctly?
- What if **extra irrelevant clauses** are added to the question? Can the machine identify and discard them to reach the correct answer?

GSM8K and GSM-Symbolic

- **GSM-Symbolic** is an extension of **GSM8K**, where questions are divided into 50 test sets, each containing 100 questions generated from 100 templates

GSM8K (Original)	GSM-Symbolic (Generated)
Benny saw a **10**-ft shark with **2 6**-inch remoras. What % of the shark's length is the combined remora length?	Olivia saw a **210**-ft whale with **7 72**-inch remoras. What % of the whale's length is the combined remora length?
Answer: **10%**	Answer: **20%**

- **GSM-P1** and **GSM-P2** are built on GSM-Symbolic by adding one and two extra irrelevant clauses respectively

GSM-Symbolic (Base)	GSM-P1 (One Extra Clause)
Olivia saw a **210**-ft whale with **7 72**-inch remoras. What % of the whale's length is the combined remora length?	Olivia saw a **210**-ft whale with **7 72**-inch remoras. The whale was swimming alongside an **85**-ft dolphin. What % of the whale's length is the combined remora length?
Answer: **20%**	Answer: **20%** (extra clause irrelevant)

Formal Specification

- Every problem in natural language can be translated into a **formal specification**
- A human reasoner performs this translation implicitly: they **comprehend** the question and apply **logical inference** to derive the answer
- But can a machine do the same?

Natural Language	Formal Specification

Mei saw a 90-ft dolphin with 3 36-inch remoras. What % of the dolphin's length is the combined remora length?	`#init:`
	`dolphin = 90 ft, remora = 36 in, n = 3`
	`#cond:`
	`ans = (n × remora) / (dolphin × 12) × 100`
	`#answer: ans = 10%`

Methodology

Model: GPT-4o-mini · 8-shot · greedy decoding

- **Baseline** — standard 8-shot CoT from GSM8K; model solves the question directly
- **Formal w/ template** — shots from GSM-Symbolic each with formal spec attached; model must **derive its own formal spec** for the target question before solving
- **Formal w/o template** — same shots as baseline with no formal spec shown; model must **derive its own formal spec** from instruction alone before solving

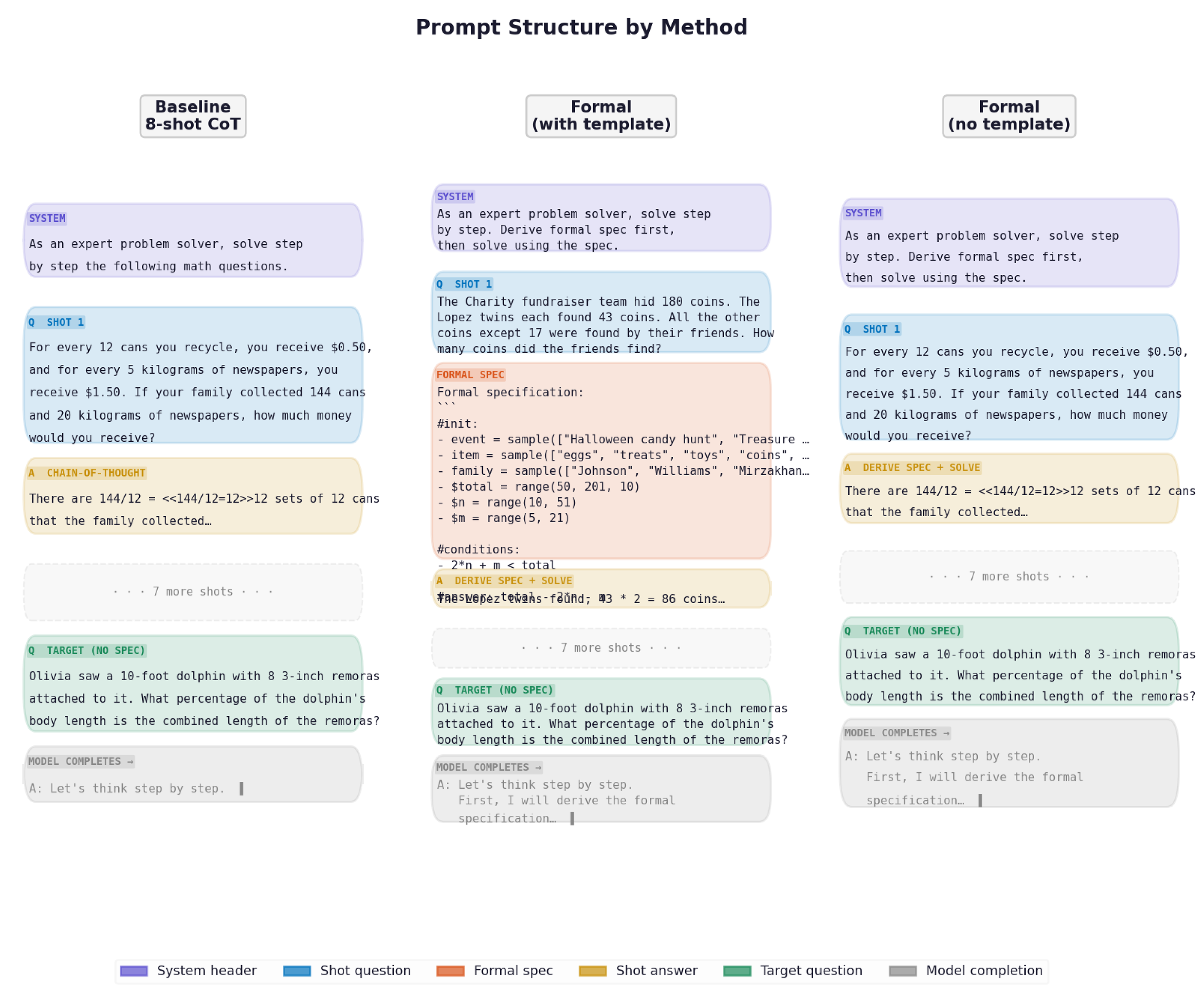

Figure 1. Prompt structure comparison across all three methods.

Experimental Result

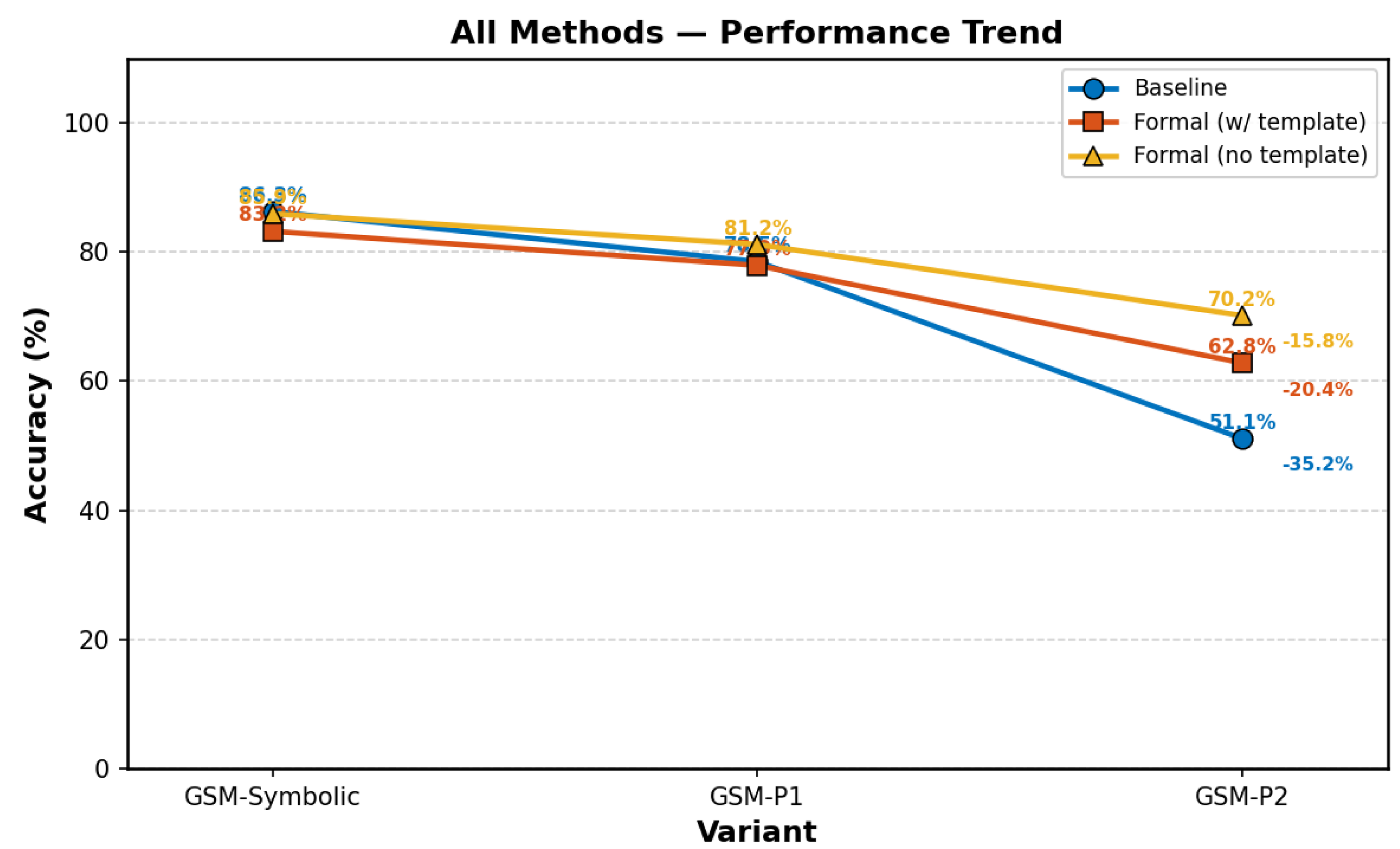

Figure 2. Accuracy across variants for all three methods. Each point is mean accuracy across instance sets.

Analysis

Model	GSM8K (FULL)	GSM-Symbolic	GSM-P1	GSM-P2
GPT-4o	95.2	94.9±1.87	93.9±2.59	88.0±3.43
GPT-4o-mini	94.2	91.7±2.02	81.1±3.05	72.4±4.57

Table 3: Results from Mirzadeh et al. (2025) Tab. 1. All values are accuracy (%). Mean ± std computed across 50 instance sets.

Method	Variant	Mean	Std	Min	Max	N
Baseline	GSM-Symbolic	86.3	2.3	81.0	90.0	50
	GSM-P1	78.5	2.5	73.0	83.0	50
	GSM-P2	51.1	4.7	40.0	64.0	50
Formal (w/ template)	GSM-Symbolic	83.2	1.8	80.0	87.0	25
	GSM-P1	77.9	2.8	70.0	83.0	25
	GSM-P2	62.8	5.0	54.0	72.0	25
Formal (no template)	GSM-Symbolic	85.9	2.2	81.0	89.0	25
	GSM-P1	81.2	2.6	76.0	86.0	25
	GSM-P2	70.2	3.1	62.0	76.0	25

Table 4: Our results using `gpt-4o-mini`, 8-shot CoT, greedy decoding. Mean ± std computed across N instance sets. All values are accuracy (%).

Baseline replication. Performance degrades consistently as variant difficulty increases, confirming the findings of Mirzadeh et al. The ∼20% gap on GSM-P2 relative to the paper is attributed to the 512-token output limit causing truncation on longer reasoning chains.

Formal specification. Both formal experiments show a smaller drop from GSM-Symbolic to GSM-P1 (∼5%) compared to the baseline (∼8%), suggesting formalisation partially helps the model focus on relevant variables. Two explanations:

- The formalisation step genuinely improves reasoning capacity
- The larger token budget (1024 vs 512) provides more computation time for reasoning

GSM-P2. Both formal experiments show little improvement, indicating that two irrelevant clauses remain challenging regardless of the prompting strategy employed.

Conclusion

LLMs exhibit **persistent limitations** in mathematical reasoning that are not easily remedied by prompt engineering alone. Formal specification injection offers modest improvement on moderately harder variants but provides **little benefit on GSM-P2**, pointing toward deeper architectural or training-level factors as the root cause.

Further Thoughts

- Turing proposed three components for imitating an adult human mind:
  (a) The **initial state** of the mind, say at birth
  (b) The **education** to which it has been subjected
  (c) Other **experience** not to be described as education, to which it has been subject.
- He then proposed that, "instead of trying to produce a **programme** to simulate the **adult's** mind, why not rather try to produce one which simulates the **child's**?"
- The modern LLM training pipeline [Asperti et al., 2026]:
  (1) Pre-training
  (2) Supervised Fine-Tuning (SFT)
  (3) Reward Model Training
  (4) Reinforcement Learning (RL)
  (5) Optional additional SFT
- Stages (2)–(5) correspond to Turing's (b) and (c) — **education and experience**
- Open question: is **pre-training** equivalent to Turing's **initial state**? And how should the "initial state" of a machine be determined?

References

[1] Turing, A.M. *Computing Machinery and Intelligence*. Mind, 1950.
[2] Mirzadeh et al. *GSM-Symbolic*. ICLR 2025.
[3] Asperti et al. *Thinking Machines*. BDCC 2026.
[4] Wei et al. *Chain-of-Thought Prompting*. NeurIPS 2022.

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